PCA SUMMARY — EMPLOYEE WORKFORCE SEGMENTATION

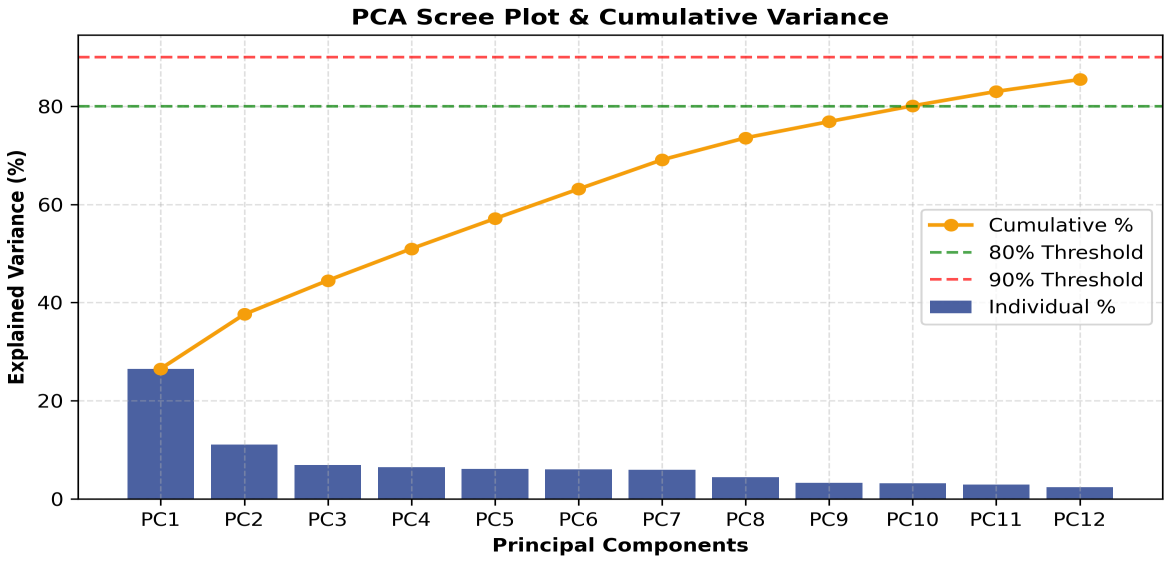
Dimensionality Reduction, Eigenvalue Decomposition & Feature Loadings Deliverable

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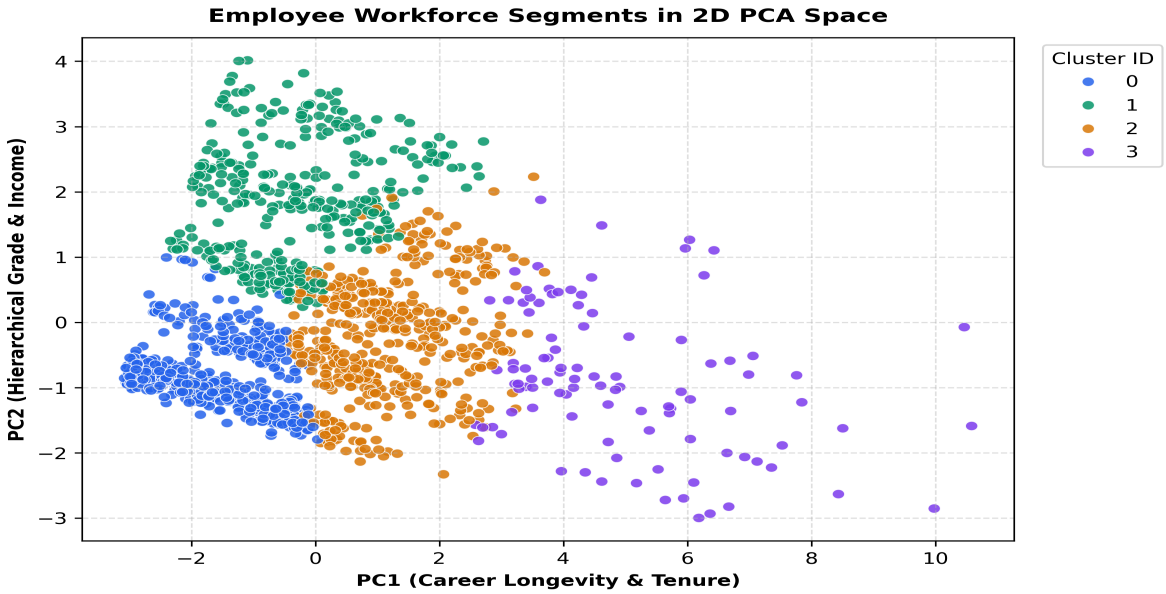
1. Dimensionality Reduction Objective:
Principal Component Analysis was executed on the 35 standardized workforce features to address the curse of dimensionality, resolve multi-collinearity, and project multi-dimensional employee profiles onto an interpretable 2D subspace for visual verification.

2. Variance Explained Summary:
- **PC1 Explained Variance:** 26.53%
- **PC2 Explained Variance:** 11.08%
- **PC1 + PC2 Combined 2D Variance:** 37.61%
- **Components for 80% Variance:** 10 components
- **Components for 90% Variance:** 15 components
- **Components for 95% Variance:** 19 components

3. Scree Plot & Cumulative Explained Variance



4. 2D PCA Workforce Projection



5. Principal Component Loadings & Interpretation

Principal component loadings represent the correlation coefficients between the original scaled features and each component. High positive or negative values identify the primary drivers of workforce variation.

Leading Contributors to PC1 (Career Longevity & Tenure):

Feature	Loading	Absolute Loading	Direction
YearsAtCompany	0.44	0.44	Positive
TotalWorkingYears	0.42	0.42	Positive
YearsSinceLastPromotion	0.37	0.37	Positive
YearsInCurrentRole	0.36	0.36	Positive
YearsWithCurrManager	0.36	0.36	Positive
Age	0.34	0.34	Positive

Leading Contributors to PC2 (Hierarchical Grade & Income):

Feature	Loading	Absolute Loading	Direction
JobLevel	0.66	0.66	Positive
MonthlyIncome	0.63	0.63	Positive
YearsSinceLastPromotion	-0.18	0.18	Negative
YearsAtCompany	-0.17	0.17	Negative
YearsInCurrentRole	-0.16	0.16	Negative
YearsWithCurrManager	-0.15	0.15	Negative

6. Information Loss & Technical Limitations

Information Loss: *While the 2D projection retains 37.6% of variance, approximately 62.4% of total feature variance is not captured in a two-dimensional visual plane. Consequently, visual proximity in 2D space does not strictly guarantee high similarity across all 35 features. For this reason, the clustering models operate on the full standardized feature matrix rather than truncating to 2 components.*

Linearity Assumption: *PCA assumes linear combinations of variables. Non-linear relationships between tenure and satisfaction may require supplementary non-linear manifold learning (e.g., UMAP or t-SNE) for fine-grained localized cluster separation.*